

Laporan Dokumentasi Deployment

Dokumen ini menjelaskan arsitektur dan konfigurasi *deployment* aplikasi "Wondr Desktop" ke lingkungan produksi, yang menggunakan Google Kubernetes Engine (GKE) dan layanan Google Cloud lainnya.

1. Arsitektur Aplikasi

Aplikasi "Wondr Desktop" terdiri dari empat komponen utama yang terpisah dalam direktori *root* proyek:

- **backend-secure-onboarding-system**: Layanan *backend* utama.
- **frontend-secure-onboarding-system**: Aplikasi *frontend* berbasis web.
- **ops-secure-onboarding-system**: Direktori untuk konfigurasi operasional.
- **verificator-secure-onboarding-system**: Layanan verifikasi pihak ketiga.

```
.
├── backend-secure-onboarding-system
├── frontend-secure-onboarding-system
├── ops-secure-onboarding-system
└── verificator-secure-onboarding-system
```

1.1 Struktur Backend & Frontend

Struktur folder **backend** menunjukkan arsitektur Spring Boot dengan fokus pada layanan otentikasi, registrasi, dan verifikasi.

```
.
├── Dockerfile
├── alert-rules.yml
├── alertmanager.yml
├── mvnw
├── pom.xml
├── prometheus.yml
└── src
    ├── main
    │   ├── java
    │   │   ├── com
    │   │   │   ├── reg
    │   │   │   │   ├── regis
    │   │   │   │   │   ├── RegistrationAbsoluteApplication.java
    │   │   │   │   │   ├── client
    │   │   │   │   │   │   ├── DukcapilWebClient.java
    │   │   │   │   │   ├── config
    │   │   │   │   │   │   ├── DukcapilClientConfig.java
    │   │   │   │   │   │   ├── OtpConfig.java
    │   │   │   │   │   │   ├── RateLimitConfig.java
    │   │   │   │   │   │   ├── SecurityConfig.java
    │   │   │   │   │   │   └── SwaggerConfig.java
```

```
├── WebConfig.java
├── controller
│   ├── LoginController.java
│   ├── OtpController.java
│   ├── RegistrationController.java
│   └── VerificationController.java
├── dto
│   ├── request
│   │   ├── DukcapilRequestDto.java
│   │   ├── EmailVerificationRequest.java
│   │   ├── NikVerificationRequest.java
│   │   ├── PhoneVerificationRequest.java
│   │   └── RegistrationRequest.java
│   └── response
│       ├── DukcapilResponseDto.java
│       ├── RegistrationResponse.java
│       └── VerificationResponse.java
├── model
│   ├── Alamat.java
│   ├── Customer.java
│   └── Wali.java
├── repository
│   └── CustomerRepository.java
├── security
│   ├── JwtAuthFilter.java
│   ├── JwtUtil.java
│   └── SecurityUtil.java
├── service
│   ├── CustomerUserDetailsService.java
│   ├── DukcapilClientService.java
│   ├── LoginAttemptService.java
│   ├── RegistrationService.java
│   └── VerificationService.java
├── resources
│   └── application.properties
└── test
    ├── java
    │   ├── com
    │   │   ├── reg
    │   │   │   ├── regis
    │   │   │   │   ├── TestSummary.md
    │   │   │   │   ├── controller
    │   │   │   │   │   ├── LoginControllerTest.java
    │   │   │   │   │   ├── RegistrationControllerTest.java
    │   │   │   │   │   └── VerificationControllerTest.java
    │   │   │   │   ├── dto
    │   │   │   │   │   ├── request
    │   │   │   │   │   │   ├── DukcapilRequestDtoTest.java
    │   │   │   │   │   │   ├── EmailVerificationRequestTest.java
    │   │   │   │   │   │   ├── NikVerificationRequestTest.java
    │   │   │   │   │   │   ├── PhoneVerificationRequestTest.java
    │   │   │   │   │   │   └── RegistrationRequestTest.java
    │   │   │   │   │   └── response
    │   │   │   │   │       └── DukcapilResponseDtoTest.java
```

```
├── RegistrationResponseTest.java
├── VerificationResponseTest.java
├── model
│   ├── AlamatTest.java
│   ├── CustomerTest.java
│   └── WaliTest.java
├── repository
│   └── CustomerRepositoryTest.java
├── security
│   ├── JwtAuthFilterTest.java
│   ├── JwtUtilTest.java
│   └── SecurityUtilTest.java
├── service
│   ├── CustomerUserDetailsServiceTest.java
│   ├── DukcapilClientServiceTest.java
│   ├── LoginAttemptServiceTest.java
│   ├── RegistrationServiceTest.java
│   └── VerificationServiceTest.java
└── resources
    └── application-test.properties
```

Struktur folder **frontend** menggunakan React dengan Vite. Konfigurasi Nginx disertakan untuk melayani aplikasi statis.

```
.
├── Dockerfile
├── eslint.config.js
├── index.html
├── nginx
│   └── default.conf
├── package-lock.json
├── package.json
├── public
│   ├── index.html
│   ├── vite.svg
│   └── wondr-logo.png
├── src
│   ├── App.jsx
│   ├── Dashboard
│   │   ├── Dashboard.jsx
│   │   └── Dashboard.module.css
│   ├── Login
│   │   ├── LoginForm.jsx
│   │   └── LoginForm.module.css
│   ├── assets
│   │   ├── Dashboard.png
│   │   ├── ...
│   │   └── wondr.png
│   ├── components
│   │   ├── PasswordForm.jsx
│   │   └── ...
```

```
├── undang.pdf
├── context
│   ├── RegisterContext.jsx
│   └── formContext.jsx
├── firebase.js
├── index.css
├── main.jsx
├── pages
│   ├── AccountConfirmation.jsx
│   ├── AddressInputPage.jsx
│   ├── CreatePassword.css
│   ├── CreatePasswordPage.jsx
│   ├── EKTPVerification.css
│   ├── EKTPVerificationPage.jsx
│   ├── EmailInput.css
│   ├── Home.css
│   ├── Home.jsx
│   ├── JenisKartuPage.css
│   ├── JenisKartuPage.jsx
│   ├── JenisTabunganPage.css
│   ├── JenisTabunganPage.jsx
│   ├── JumlahGaji.jsx
│   ├── MotherName.css
│   ├── MotherNamePage.jsx
│   ├── OccupationPage.jsx
│   ├── PenghasilanPage.jsx
│   ├── PersonalDataForm.css
│   ├── PersonalDataForm.jsx
│   ├── PhoneInputPage.css
│   ├── PhoneInputPage.jsx
│   ├── WaliIdentity.css
│   ├── WondrLanding.css
│   ├── WondrLanding.jsx
│   ├── address.css
│   ├── emailInputPage.jsx
│   ├── nameInput.css
│   ├── nameInputPage.jsx
│   ├── phoneOtpInputPage.css
│   ├── phoneOtpInputPage.jsx
│   ├── termsCondition.jsx
│   ├── termsConditions.module.css
│   ├── tujuanPembukaanRekening.jsx
│   ├── undang.jsx
│   ├── undang.module.css
│   ├── waliFormPages.css
│   ├── waliIdentityPage.jsx
│   └── waliPage.jsx
├── utils
│   ├── sanitize.js
│   └── validation.js
├── vite.config.js
└── wondr-logo.png
```

2. Infrastruktur Deployment

Aplikasi ini di-deploy menggunakan kombinasi GKE, Cloud Run, dan Cloud SQL untuk memastikan skalabilitas, efisiensi, dan manajemen yang mudah.

2.1 Google Kubernetes Engine (GKE)

- **Nama Cluster:** `gke-secure-onboarding-system`
- **Spesifikasi:**
 - **Tipe:** `Autopilot` (secara otomatis mengelola node dan pods).
 - **Versi:** `1.28.6-gke.1130000`.
 - **Lokasi:** `asia-southeast1-a` (Jakarta).
- **Deskripsi:** Cluster GKE digunakan untuk menampung `deployment backend-secure-onboarding-system` dan `frontend-secure-onboarding-system` serta database PostgreSQL.

2.2 Komunikasi Antar-Komponen dalam Cluster

- **Namespace:** *Deployment* dipisahkan ke dalam *namespace* `frontend` dan `backend`. Database PostgreSQL berada di *namespace* `backend`.
- **Network Policy:** Komunikasi antar *namespace* diatur dengan `NetworkPolicy`. Berikut adalah kebijakan yang memungkinkan *traffic* dari *namespace* `frontend` ke *service backend* dengan port `8080`.

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-frontend-to-backend
  namespace: backend
spec:
  podSelector:
    matchLabels:
      app: backend
  policyTypes:
    - Ingress
  ingress:
    - from:
        - namespaceSelector:
            matchLabels:
              name: frontend-app
          podSelector:
            matchLabels:
              app: frontend-app
      ports:
        - protocol: TCP
          port: 8080
```

2.3 Deployment & Service Kubernetes

```
k8s
├── application
│   ├── backend-deployment.yaml
│   ├── backend-networkpolicy.yaml
│   ├── database-setup.sql
│   ├── frontend-certificate.yaml
│   ├── frontend-configmap.yaml
│   ├── frontend-deployment.yaml
│   ├── frontend-service.yaml
│   ├── ingress.yaml
│   ├── postgresql-deployment.yaml
│   ├── postgresql-secret.yaml
│   └── postgresql-service.yaml
└── monitoring
    ├── alert-deployment.yaml
    ├── alert-rules.yaml
    ├── alertmanager.yaml
    ├── grafana-configmap.yaml
    ├── grafana-service.yaml
    ├── grafana-statefulset.yaml
    ├── ingress.yaml
    ├── monitoring-certificate.yaml
    ├── prome-configmap.yaml
    ├── prome-daemonset.yaml
    └── prome-service.yaml
```

a. Backend Deployment

- **Nama:** `backend-secure-onboarding-system-deployment`.
- **Replicas:** 2.
- **Image:** `gcr.io/model-parsec-465503-p3/backend-secure-onboarding-system:<IMAGE_TAG>`.
- **Service:** `backend-service` (Type: `ClusterIP`) yang mengekspos port `8080` untuk komunikasi internal.

b. Frontend Deployment

- **Nama:** `frontend-secure-onboarding-system-deployment`.
- **Replicas:** 2.
- **Image:** `gcr.io/model-parsec-465503-p3/frontend-secure-onboarding-system:<IMAGE_TAG>`.
- **Service:** `frontend-service` (Type: `NodePort`) yang mengekspos port `80` untuk diakses melalui Ingress.

c. Database Deployment

- **Nama:** `postgresql-deployment`.
- **Image:** `postgres:17-alpine`.
- **Service:** `postgresql-service` (Type: `ClusterIP`) untuk akses internal dari *backend*.

2.4 Ingress

- **Nama:** `secure-onboarding-ingress`.
- **Namespace:** `frontend`.
- **Konfigurasi:** Menggunakan Ingress GCE dengan IP statis global bernama `secure-onboarding-ip`.

- **Tujuan:** Meneruskan semua *traffic* dari domain `wondrdesktop.my.id` ke `frontend-service` pada port `80`.

2.5 Layanan Eksternal

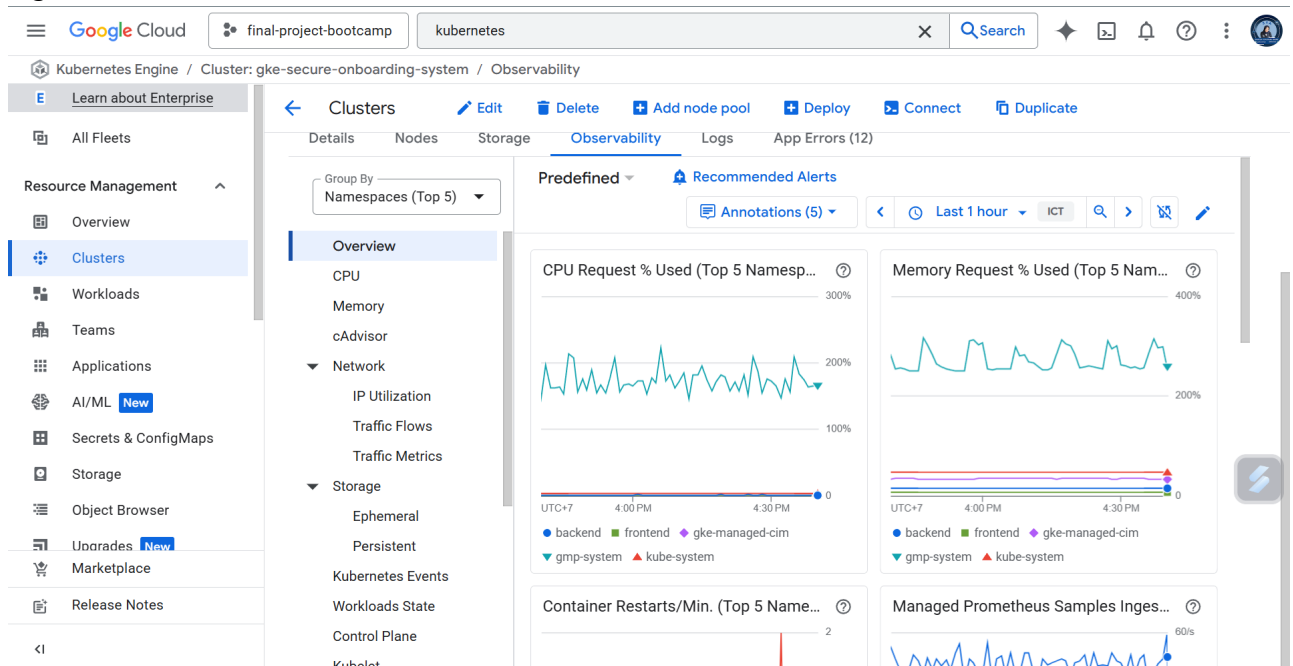
Aplikasi ini menggunakan layanan eksternal untuk fungsionalitas tertentu:

- **Verifikasi OTP SMS:** Menggunakan layanan dari **Firebase Auth**.
- **Verifikator Dukcapil:** Layanan ini di-*deploy* sebagai **Cloud Run Service** dengan **Cloud SQL** sebagai databasenya. Hal ini dilakukan untuk memisahkan layanan yang spesifik dari *cluster* utama, memastikan skalabilitas yang lebih baik untuk layanan yang sifatnya terisolasi.
 - **Nama Layanan:** `verificator-secure-onboarding-system`
 - **Endpoint:** `https://verificator-secure-onboarding-system-441501015598.asia-southeast1.run.app`

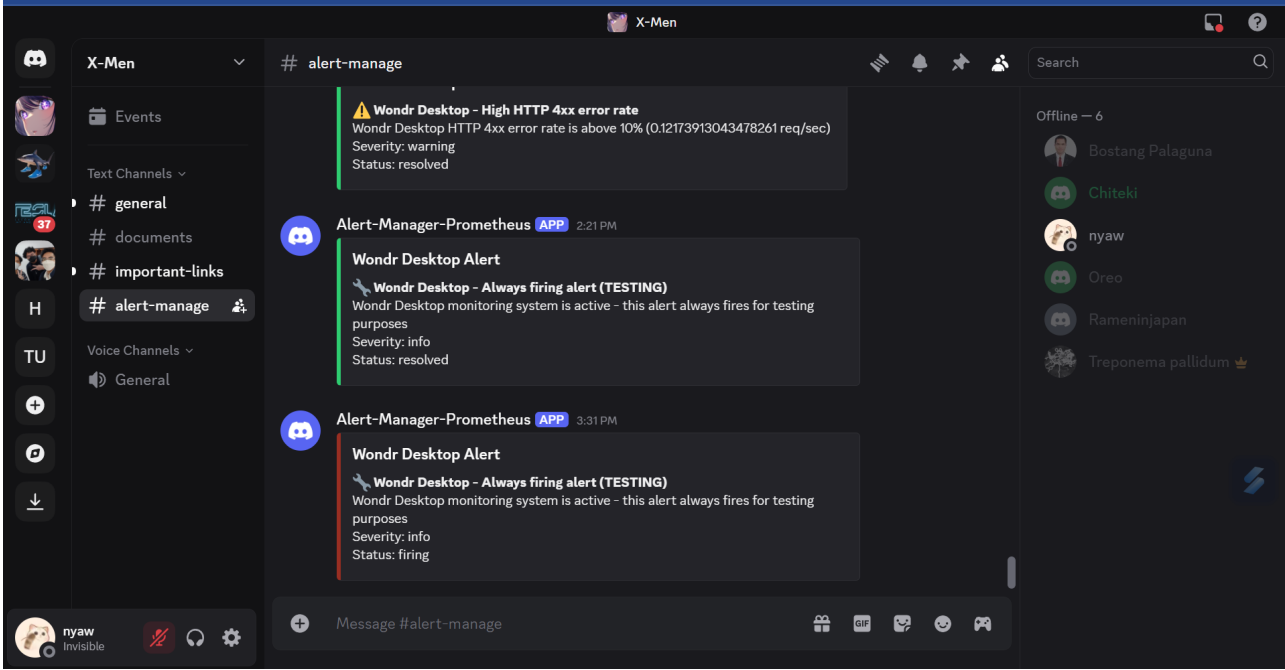
2.6 Monitoring Menggunakan Prometheus, Grafana, Google Kubernetes Observability, dan AlertManager

Monitoring dilakukan mendapatkan visibilitas menyeluruh terhadap sistem yang berjalan pada cluster k8s.

- Prometheus digunakan untuk mengumpulkan dan menyimpan metrics (<https://prometheus.wondrdesktop.my.id/query>)
- Grafana digunakan untuk menampilkan dashboard visualisasi data metrics (<https://grafana.wondrdesktop.my.id/>)
- Google Kubernetes Observability digunakan untuk menampilkan performance dari cluster yang digunakan.



- Alert Manager digunakan untuk memberikan alert melalui channel Discord jika terdapat anomali pada aplikasi.



Informasi Detail

informasi detail dari gke-cluster :

gke-secure-onboarding-system

 Automatic upgrades and other Google maintenance tasks may run at any time, increasing transient disruptions to your workloads. Set window.

[View details](#)

Details	Nodes	Storage	Observability	Logs	App Errors (10)
Cluster basics					
Name	gke-secure-onboarding-system				
Tier 	Standard				
Mode 	Standard				
Location type	Zonal				
Control plane zone	asia-southeast1-a				
Default node zones 	asia-southeast1-a				
Total size	4				
Release channel 	Regular channel				
Version 	1.33.2-gke.1111000				 Upgrade available
Current COS version	cos-121-18867-90-62				
End of standard support 	Aug 3, 2026				
End of extended support 	Jun 3, 2027				
Rollout sequence 	To use rollout sequencing, register your cluster to a fleet				

informasi backend-deployment:

✔ backend-deployment

Overview	Details	Observability	Revision history	Events
Cluster	gke-secure-onboarding-system			
Namespace	backend			
Created	Aug 3, 2025, 9:44:45 AM			
Labels	No labels set			
Annotations	deployment.kubernetes.io/revision: 3 ▼ Show all annotations			
Replicas	1 updated, 1 ready, 1 available, 0 unavailable			
Label selector	app = backend			
Update strategy ?	Rolling update, Max unavailable: 25%, Max surge: 25%			
Min time ready before available	0 s			
Progress deadline	600 s			
Revision history limit	10			

informasi frontend-deployment:

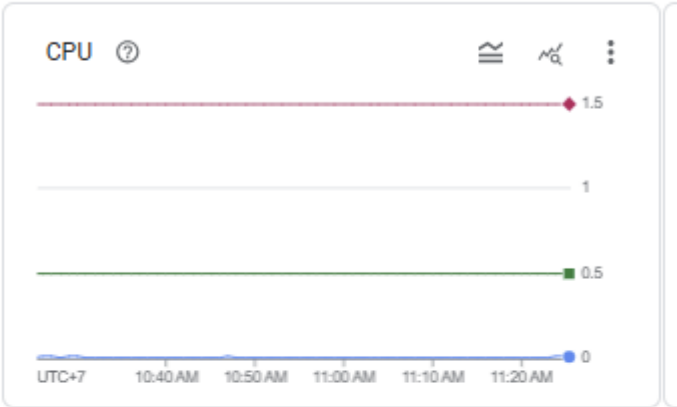
✔ frontend-deployment

Overview	Details	Observability	Revision history	Events	Logs	App Errors (0)	YAML
Cluster	gke-secure-onboarding-system						
Namespace	frontend						
Created	Aug 3, 2025, 12:12:17 PM						
Labels	app: frontend-app						
Annotations	deployment.kubernetes.io/revision: 2 ▼ Show all annotations						
Replicas	1 updated, 1 ready, 1 available, 0 unavailable						
Label selector	app = frontend-app						
Update strategy ?	Rolling update, Max unavailable: 25%, Max surge: 25%						
Min time ready before available	0 s						
Progress deadline	600 s						
Revision history limit	10						

informasi postgresql-deployment:

✓ postgresql

Details Events Logs YAML



Cluster	gke-secure-onboarding-system
Namespace	backend
Created	Jul 30, 2025, 1:57:30 PM
Labels	app: postgresql
Annotations	Show annotations
Logs ?	Container logs , Audit logs
Label selector	app = postgresql
Replicas	1
Observed generations	1
Service	postgresql-service
Persistent volume claims specification	postgres-data

informasi backend-service:

✓ backend-service

Overview Details Events Logs App Errors (0) YAML

Cluster	gke-secure-onboarding-system
Namespace	backend
Created	Aug 3, 2025, 9:44:45 AM
Labels	No labels set
Annotations	cloud.google.com/neg: {"ingress":true} Show all annotations
Label selector	app = backend
Pods	1 current / 1 desired
Type	ClusterIP

Cluster IP

Cluster IP	34.118.236.255
------------	----------------

informasi frontend-service:

✔ frontend-service

Overview	Details	Events	Logs	App Errors (0)	YAML
Cluster	gke-secure-onboarding-system				
Namespace	frontend				
Created	Jul 30, 2025, 4:13:27 PM				
Labels	No labels set				
Annotations	cloud.google.com/neg: {"ingress":true} cloud.google.com/neg-status: {"network_endpoint_groups":{"80":"k8s1-3963eb42-frontend-frontend-service-80-ed8857e4"},"zones":["asia-southeast1-a"]} ▼ Show all annotations				
Label selector	app = frontend-app				
Pods	1 current / 1 desired				
Type	ClusterIP				
Cluster IP					
Cluster IP	34.118.230.170				

informasi postgresql-service:

✔ postgresql-service

Overview	Details	Events	Logs	App Errors (0)	YAML
Cluster	gke-secure-onboarding-system				
Namespace	backend				
Created	Jul 30, 2025, 1:57:50 PM				
Labels	app: postgresql				
Annotations	cloud.google.com/neg: {"ingress":true} ▼ Show all annotations				
Label selector	app = postgresql				
Pods	1 current / 1 desired				
Type	ClusterIP				
Cluster IP					
Cluster IP	34.118.239.149				

informasi secure-onboarding-ingress:

secure-onboarding-ingress

DetailsTelemetryLogsEventsYAML

Cluster

gke-secure-onboarding-system

Namespace

frontend

Created

Aug 3, 2025, 12:31:22 PM

Labels

No labels set

Annotations

ingress.kubernetes.io/backends: {"k8s1-3963eb42-frontend-frontend-service-80-ed8857e4":"HEALTHY", "k8s1-3963eb42-kube-system-default-http-backend-80-be83669c":"HEALTHY"}
ingress.kubernetes.io/forwarding-rule: k8s2-fr-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy
ingress.kubernetes.io/target-proxy: k8s2-tp-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy
ingress.kubernetes.io/url-map: k8s2-um-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy
kubernetes.io/ingress.class: gce
kubernetes.io/ingress.global-static-ip-name: secure-onboarding-ip

Show all annotations

Type

Ingress

Load balancer

External HTTP(S) LB

IP address

34.49.249.12

Routes

Route	Service	Pod selector	Clusters	Pods
wondrdesktop.my.id/*	frontend-service	app = frontend-app	gke-secure-onboarding-system	1/1

Serving pods

Service	Name	Status	Restarts	Created on
frontend-service	frontend-deployment-69fb7bc8c7-6jcpq	Running	0	Aug 4, 2025, 11:21:07 AM

Load balancing resources

Load balancer	k8s2-um-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy
Target proxies	k8s2-tp-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy
Forwarding rules	k8s2-fr-ge1n6rz1-frontend-secure-onboarding-ingress-m6uyswvy
Backend services	k8s1-3963eb42-frontend-frontend-service-80-ed8857e4 k8s1-3963eb42-kube-system-default-http-backend-80-be03669c

informasi service **verificator-secure-onboarding-system** :

verificator-secure-onboarding-system

Region: asia-southeast1

URL: https://verificator-secure-onboarding-system-441501015598.asia-southeast1.run.app

MetricsSLOsLogsRevisionsTriggersNetworkingSecurityYAML

Revisions

Manage traffic

Filter

Filter revisions

Name	Traffic	D	Actions
verificator-secure-onboarding-system-00019-njt	100% (to latest)	2	
verificator-secure-onboarding-system-00018-cg9	0%	1	
verificator-secure-onboarding-system-00017-mpf	0%	4	
verificator-secure-onboarding-system-00016-fpz	0%	4	
verificator-secure-onboarding-system-00012-47t	0%	4	
verificator-secure-onboarding-system-00008-r2k	0%	6	

verificator-secure-onboarding-system-00019-njt

Deployed by bostangsteiitb2020@gmail.com using Cloud Console

Containers

Volumes

Networking

Security

YAML

General

Billing	Request-based
Startup CPU boost	Enabled
Concurrency	80
Request timeout	300 seconds
Execution environment	Default

Autoscaling

Revision max instances	100
------------------------	-----

Image	asia.gcr.io/primeval-rune-467212-t9/verificator-sec...
Port	8080
Build	(no build information available)
Source	(no source information available)
Command and args	(container endpoint)
CPU limit	1
Memory limit	512MiB

informasi **wondr-desktop-otp**:

 **Firebase**

Project Overview



Project shortcuts

Authentication

What's new

AI Logic

NEW

Product categories

Build

Run

Analytics

AI

Related development tools

 [Firebase Studio](#)

wondr-desktop-otp

Project settings

General

Cloud Messaging

Integrations

Service accounts

Data privacy

Users and permissions

Alerts

Your project

Project name

wondr-desktop-otp



Project ID

wondr-desktop-otp



Project number

15939231912



Web API Key

my4j6_hPQR_7eFyo



Environment

This setting customizes your project for different stages of the app lifecycle

Environment type

Unspecified

